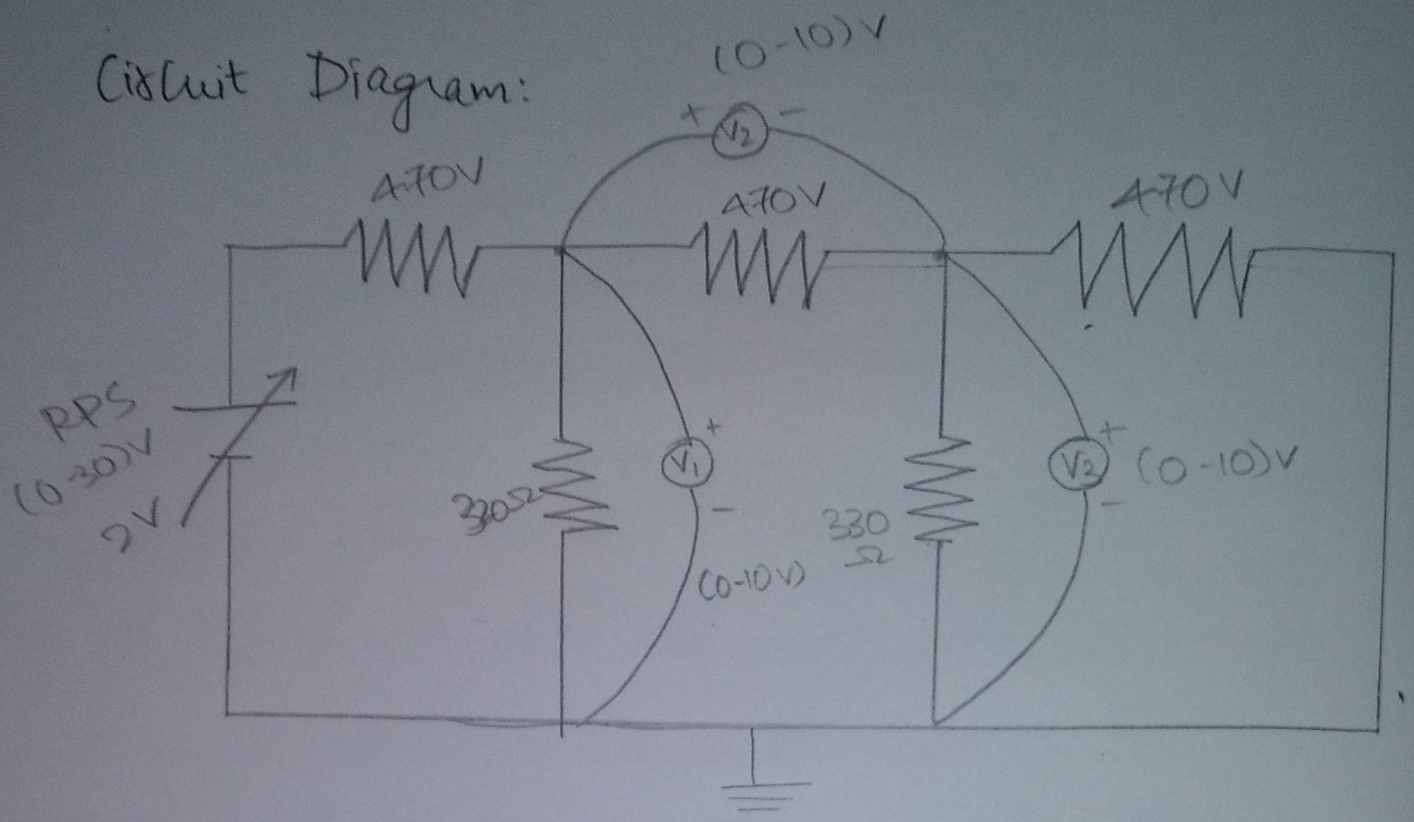
Circuit Diagram:



Tabular column

Parameter	Theoretical	Practical
Supply voltage	2V	2V
Node voltage (VA)	0.64V	0.62V
Node voltage (VB)	0.19V	0.20V
Voltage b/w A & B (V_{AB})	0.45V	0.42V

S/P verified
10/2/21

EXP.NO.: 3 Date: 10/2/26	DETERMINATION OF VOLTAGE IN CIRCUIT USING NODAL ANALYSIS

AIM:

To determine the Voltage in the circuit using nodal analysis both theoretically and practically for a given DC circuit

APPARATUS REQUIRED:

SL.NO	APPARATUS	SPECIFICATIONS	QUANTITY
1	Regulated Power Supply (RPS)	(0-30 V)	1
2	Multimeter	–	1
3	Resistors	470Ω, 330Ω	3, 2
4	Bread Board	–	1

PROCEDURE:

1. Give connections as per the circuit diagram
2. Switch ON the supply, vary the RPS and set a particular input voltage
3. Note down the readings of ammeters and voltmeters and tabulate them
4. Vary the RPS for different input voltages and

Calculations:

Node A

$$\frac{V_A - 2}{470} + \frac{V_A}{330} + \frac{V_A - V_B}{470} = 0$$

$$33(V_A - 2) + 47V_A + 33(V_A + V_B) = 0$$

$$113V_A - 33V_B = 66 \quad \text{— (1)}$$

Node B

$$\frac{V_B - V_A}{470} + \frac{V_B}{330} + \frac{V_B}{470} = 0$$

$$33(V_B - V_A) + 47V_B + 33V_B = 0 \quad \text{— (2)}$$

(1) & (2) $\Rightarrow$ $V_B = 0.19V$ $\quad V_{AB} = 0.45V$

$V_{BA} = 0.64V$

note down the readings of all the meters.

5. Reduce the RPS to its minimum value and switch OFF the supply

6. Using the tabulated values, verify kirchoff's laws Practically, and verify it theoritically.

MARKS SPLIT UP

SL NO	DESCRIPTION	MARKS ALLOTED	MARKS OBTAINED
1	Experimental setup/ Connections	20	20
2	Observations/ Readings	20	20
3	Calculations/Results	20	20
4	Viva	20	20
5	Record	20	20
		Total	100

RESULT:

10

Thus the nodal analysis is verified practically & theoretically. The resultant voltages for 0.45 V supply are: a) The voltage V1 is 0.64V

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b) The voltage V2 is 0.19 V

c) The voltage V3 is 0.45V